

Problem 4, Cell 2: Sun's conjecture for $k = 4$

Official statement. For integers a and $m \geq 1$ write $a \pmod{m} = \{a + mt : t \in \mathbb{Z}\}$. A finite family of classes $a_1 \pmod{m_1}, \dots, a_k \pmod{m_k}$ is pairwise disjoint if $a_i \pmod{m_i} \cap a_j \pmod{m_j} = \emptyset$ whenever $i < j$. Nothing is assumed about the moduli beyond $m_i \geq 1$: they may repeat, and the residues a_i may repeat as well. By the Chinese remainder theorem,

$$a_i \pmod{m_i} \cap a_j \pmod{m_j} \neq \emptyset \iff \gcd(m_i, m_j) \mid a_i - a_j. \quad (*)$$

Question. Let $k \geq 2$ and let $a_1 \pmod{m_1}, \dots, a_k \pmod{m_k}$ be pairwise disjoint. Must there be a pair $i < j$ with $\gcd(m_i, m_j) \geq k$?

The same question for four classes. Prove the statement for $k = 4$.

Theorem 1 below is exactly this statement. The notation $a \pmod{m}$ below is the class $a \pmod{m}$ of the statement. Lemma 2 is (*), proved for completeness.

Known result. The question is Sun's disjoint congruence classes conjecture [2]. O'Bryant [1] proved it for all $k \leq 20$, so this case is known. The proof below is an independent proof and is self-contained.

For an integer $m \geq 1$ and $a \in \mathbb{Z}$ write $a \pmod{m} = \{x \in \mathbb{Z} : x \equiv a \pmod{m}\}$. A family of residue classes is *disjoint* if its members are pairwise disjoint.

Theorem 1. *Let $a_1 \pmod{m_1}, \dots, a_4 \pmod{m_4}$ be four pairwise disjoint residue classes. Then $\gcd(m_i, m_j) \geq 4$ for some $i \neq j$.*

1 When do two classes meet?

Lemma 2. *Let $m, n \geq 1$, $a, b \in \mathbb{Z}$ and $g = \gcd(m, n)$. Then $(a \pmod{m}) \cap (b \pmod{n}) \neq \emptyset$ if and only if $g \mid a - b$.*

Proof. ($\Rightarrow$) If $x \equiv a \pmod{m}$ and $x \equiv b \pmod{n}$, then $g \mid m$ and $g \mid n$ give $a \equiv x \equiv b \pmod{g}$.

($\Leftarrow$) By Bézout there are $u, v \in \mathbb{Z}$ with $um + vn = g$. Write $a - b = gt$ and set $x = a - umt$. Then $x \equiv a \pmod{m}$, and $x = a - (g - vn)t = b + vnt \equiv b \pmod{n}$. $\square$

Corollary 3. *If $a \pmod{m}$ and $b \pmod{n}$ are disjoint and $g = \gcd(m, n)$, then $g \geq 2$ and $a \not\equiv b \pmod{g}$.*

Proof. By Lemma 2, disjointness means $g \nmid a - b$, and this forces $g \neq 1$. $\square$

2 Proof of Theorem 1

Suppose, for contradiction, that $\gcd(m_i, m_j) \leq 3$ for all $i \neq j$. Write $g_{ij} = \gcd(m_i, m_j)$ and $[4] = \{1, 2, 3, 4\}$. By Corollary 3,

$$g_{ij} \in \{2, 3\} \quad \text{for all } i \neq j. \quad (1)$$

Let

$$E = \{i \in [4] : 2 \mid m_i\}, \quad T = \{i \in [4] : 3 \mid m_i\}.$$

Step 1: $E \cup T = [4]$ and $|E \cap T| \leq 1$. Given i , pick any $j \neq i$. By (1), g_{ij} is 2 or 3 and divides m_i , so $i \in E \cup T$. If two distinct indices i, j were both in $E \cap T$, then $6 \mid m_i$ and $6 \mid m_j$. This would give $g_{ij} \geq 6$, contradicting (1).

Step 2: every pair lies in E or in T . If $g_{ij} = 2$, then $2 \mid m_i$ and $2 \mid m_j$, so $i, j \in E$. If $g_{ij} = 3$, then $i, j \in T$.

Step 3: $E \setminus T = \emptyset$ or $T \setminus E = \emptyset$. Suppose $i \in E \setminus T$ and $j \in T \setminus E$. By Step 2, either $i, j \in E$, contradicting $j \notin E$, or $i, j \in T$, contradicting $i \notin T$.

Step 4: conclusion. By Step 3 and Step 1, one of two cases holds.

Case A: $T \setminus E = \emptyset$. Then $[4] = E \cup T = E$. If some pair had $g_{ij} = 3$, Step 2 would put $i, j \in T \cap E$. That contradicts $|E \cap T| \leq 1$ from Step 1. So by (1), $g_{ij} = 2$ for every pair $i \neq j$. Corollary 3 then gives $a_i \not\equiv a_j \pmod{2}$ for all $i \neq j$. So the four integers $a_1, \dots, a_4$ would be pairwise incongruent modulo 2, which the pigeonhole principle forbids (only $2 < 4$ classes).

Case B: $E \setminus T = \emptyset$. Then $[4] = T$. If some pair had $g_{ij} = 2$, Step 2 would put $i, j \in E \cap T$, again contradicting Step 1. So $g_{ij} = 3$ for every pair. Corollary 3 then gives that $a_1, \dots, a_4$ are pairwise incongruent modulo 3. This is impossible by pigeonhole (only $3 < 4$ classes).

Both cases are contradictory. Hence $g_{ij} \geq 4$ for some $i \neq j$. □

Remark 4. 1. The bound is sharp: $r \bmod 4$ for $r = 0, 1, 2, 3$ are disjoint and all pairwise gcds equal 4.

2. The argument does not use the density bound $\sum_i 1/m_i \leq 1$. Only the arithmetic of the primes 2 and 3 is needed. Moduli $m_i = 1$ are excluded automatically by (1).
3. *Computer sanity check, not part of the proof.* The script `check_p4.py` searched all moduli $m_1 \leq \dots \leq m_4 \leq 60$ with all pairwise gcds ≤ 3 (214 078 tuples), and all residues (with $a_1 = 0$, by translation). It found no disjoint system, which agrees with the theorem. Disjointness is checked literally, from the definition, and not via (*): the classes $a \bmod m$ and $b \bmod n$ meet iff $b - a \equiv mt \pmod{n}$ for some t , and the set $\{mt \bmod n : 0 \leq t < n\}$ is listed explicitly. For moduli ≤ 20 the search was repeated with an element-by-element comparison over one period.

References

- [1] K. O'Bryant, *On Z.-W. Sun's disjoint congruence classes conjecture*, in *Combinatorial Number Theory*, de Gruyter, Berlin, 2007, pp. 403–411; arXiv:math/0604347.

- [2] Z.-W. Sun, *Finite covers of groups by cosets or subgroups*, Internat. J. Math. **17** (2006), 1047–1064; arXiv:math/0501451.